Model Regulations for P2P Transactions of Green Energy

Presented by

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Introduction to P2P

- **How it Works**

Peer-to-Peer (P2P) energy trading enables electricity consumers and rooftop solar prosumers to directly transact renewable energy through a secure digital platform while utilizing the existing distribution network.

Instead of exporting surplus solar energy only to the DISCOM under conventional schemes, prosumers can sell electricity directly to nearby consumers at mutually agreed prices through a regulated marketplace.

The Distribution Licensee continues to operate and maintain the grid while enabling secure scheduling, metering, and settlement of P2P transactions.

Key Benefits and Technological Enablers

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Key Benefits

- Empowering Consumers with Energy Choice
- Additional Revenue for Prosumers
- Increased Adoption of Rooftop Solar
- Transparent and Verifiable Transactions
- Consumer Participation in Energy Markets
- Promotion of Clean Energy Consumption

- Enhanced Utilization of Distributed Renewable Energy
- Reduction in Technical Losses
- Improved Visibility of Distributed Energy Resources
- Optimization of Distribution Infrastructure
- Support for Grid Flexibility
- Enhanced Digitalization of Distribution Utilities

Technological Enablers

- The emergence of smart meters,
- digital transaction platforms,
- blockchain technologies,
- and advanced Meter Data Management (MDM) systems

Important Stakeholders

- **Regulatory Authorities**
  - State Electricity Regulatory Commission (SERC)
  - Forum of Regulators (FoR)

- **Distribution Utilities**
  - State Owned Power Distribution Companies

- **Technology & Platform Providers**
  - Blockchain / Digital Energy Platform Providers
  - Smart Meter Technology Providers
  - AMISPs/ Meter Data Management System Providers

- **Market Participants**
  - P2P Prosumers (Rooftop Solar Owners)
  - P2P Consumers
  - Service Providers operating P2P platforms

Supervisory & Standards Bodies

- *Central Electricity Authority (CEA)*
- *Ministry of Power (MoP)*
- *Ministry of New and Renewable Energy (MNRE)*

Core Components

Rooftop Solar Prosumers
Generate renewable electricity and inject surplus energy into the distribution grid.

P2P Platform
A blockchain-enabled or equivalent secure digital platform that:
- Matches buyers and sellers
- Records energy transactions
- Enables price discovery
- Maintains immutable transaction logs

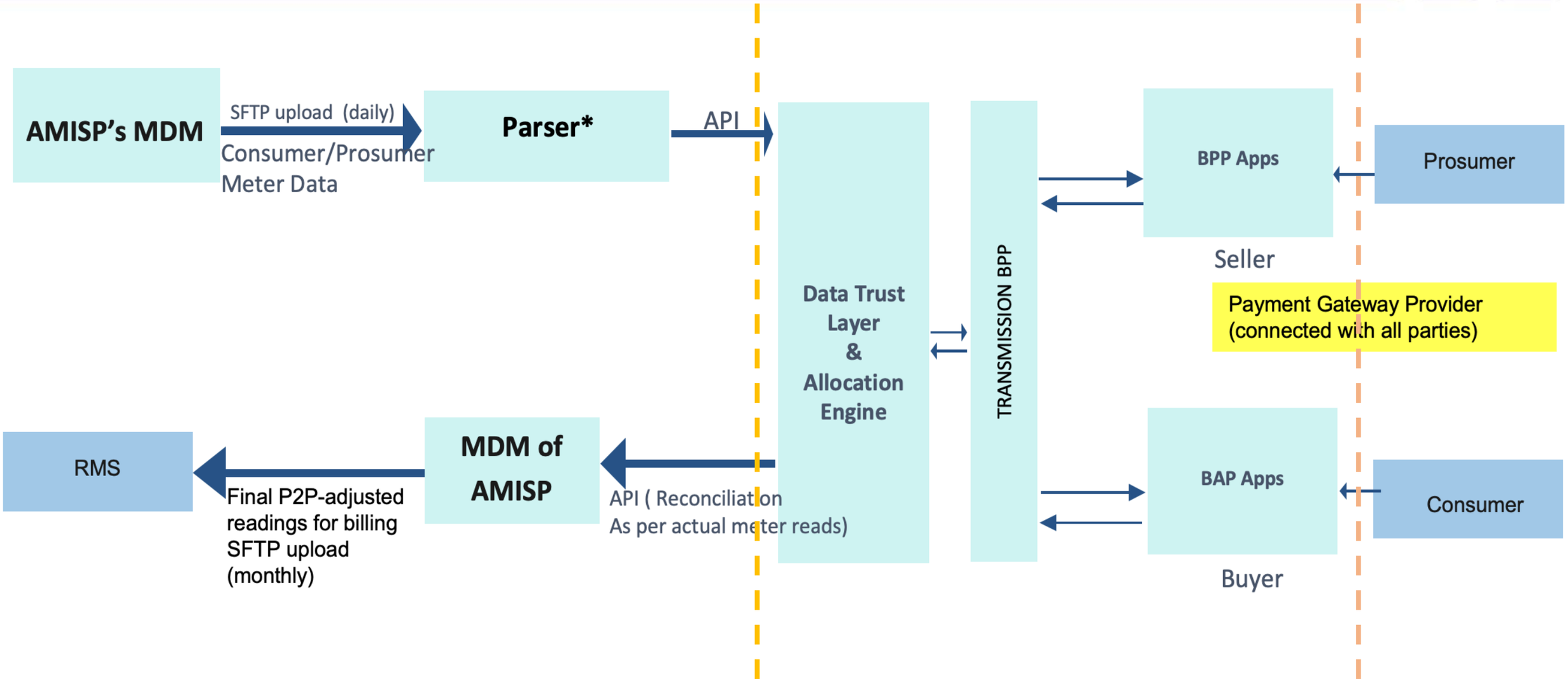
Smart Metering Infrastructure
Smart meters record energy injection and drawl in **15-minute time blocks**, enabling accurate measurement of scheduled and actual transactions.

Meter Data Management (MDM) System
The DISCOM's MDM system validates meter data and provides authenticated energy records for settlement.

Distribution Network
The DISCOM network enables physical delivery of electricity between P2P participants while maintaining grid stability and safety.

Settlement & Billing System
Validated meter data is used to calculate:
- Energy transacted between participants
- Transaction charges
- Applicable network charges

P2P Energy Trading Architecture



P2P Regulations Across India

Comparison of P2P Regulations in 4 States

Sl. No.	Parameter	Uttar Pradesh	Delhi	Karnataka	Kerala
1	Date of Issue	05 April 2023	24 June 2024	06 August 2024	05 November 2025
2	Title	Guidelines for Peer-to-Peer Solar Energy Transaction through Blockchain Platform	Delhi Electricity Regulatory Commission (Peer-to-Peer Energy Transaction) Guidelines, 2024	KERC (Implementation of Peer-to-Peer Solar Energy Transaction) Regulations, 2024	Kerala State Electricity Regulatory Commission (Renewable Energy and Related Matters) Regulations, 2025
3	Eligible Participants	Prosumers with rooftop solar (Gross/Net metering) and consumers within same DISCOM	Prosumers and consumers ≤ 200 kW sanctioned load; RE capacity up to 500% of sanctioned load	Registered SRTPV consumers having Net or Gross metering	Prosumers and consumers with RE / BESS (future rollout)
4	Registration Process	Registration with DISCOM and empanelled P2P platform	Registration through DISCOM and approved P2P platform	Registration with DISCOM and platform aggregator	Registration with DISCOM (KSEBL)
5	Registration Charge	As specified by DISCOM / platform	Nil	As specified by DISCOM	Not specified
6	Metering Requirement	Post-paid Smart Meter mandatory for both prosumer and consumer	ToD compliant energy meter or smart meter mandatory	Post-paid smart meter / ToD compliant energy meter	Smart meter / ToD meter expected
7	Technology Platform	Mandatory blockchain-based P2P platform	Blockchain or any other secure digital technology	Digital trading platform	To be notified after pilot study
8	Transaction Price	Mutually agreed price between participants on P2P platform	Dynamic / mutually agreed price between prosumers and consumers	Market-discovered / mutually agreed	To be notified

Comparison of P2P Regulations in 4 States

Sl. No.	Parameter	Uttar Pradesh	Delhi	Karnataka	Kerala
9	Transaction Charges	₹0.42/kWh (incl. GST) shared equally by buyer and seller	To be approved by DERC based on DISCOM petition	As approved under pilot	To be notified
10	Wheeling Charges	Applicable as per UPERC tariff order	Nil for both prosumers and consumers	Applicable except when connected at same DT / substation	Likely applicable (not waived)
11	Network / Open Access Charges	Wheeling charges applicable; CSS waived; Additional surcharge Nil	Complete waiver of wheeling, CSS and additional surcharge till 31 March 2027	Relaxed / waived during pilot	To be notified
12	Platform Service Fee	Payable to P2P service provider as approved by UPERC	Regulated platform service fee approved by DERC	Regulated by Commission / DISCOM	To be determined by KSERC
13	Settlement and Billing	Billing by DISCOM; P2P settlement integrated with DISCOM billing cycle	Billing by DISCOM / service provider; priority settlement for P2P energy	Settlement handled under sandbox arrangement	Energy accounting from Jan 2026; P2P after pilot
14	RPO Treatment for P2P Transaction	RPO benefit retained by DISCOM	Counts towards DISCOM RPO if consumer not obligated	RPO remains with DISCOM	RPO accounting with DISCOM
15	Notable / Distinguishing Features	First detailed state-level P2P framework in India; clear scheduling, deviation and settlement rules	Most progressive framework; strong financial incentives and charge exemptions	Pilot-oriented approach to test market design before full regulations	Explicit recognition of P2P, VPP and V2G; implementation deferred pending pilots

What we Learnt form the 4 States

- Delhi has the most progressive regulatory framework
- Regulatory maturity varies significantly across states
- Smart metering and ToD metering are foundational requirements
- Blockchain is not universally mandated despite early pilots
- Distribution utilities retain control over settlement and RPO benefits
- Market pricing mechanisms are becoming more flexible and market-driven

Model Regulation for P2P Energy Trading

Scope of Regulation

- Peer-to-Peer (P2P) Renewable Energy Transactions involving solar energy or any other Renewable Energy System approved by MNRE or UPERC.
- Prosumer-to-Consumer **energy exchange**, within the area of a single Distribution Licensee.
- **Registration,** metering, scheduling, settlement, deviation management, and commercial arrangements for P2P transactions.
- **Integration of P2P platforms with DISCOM** systems including MDM, billing, metering, and operational monitoring.
- **Technical standards,** cybersecurity, and data governance for blockchain or other approved digital transaction technologies.
- **Rights and responsibilities** of Prosumers, Consumers, Distribution Licensees, and Service Providers.
- **Transition from existing schemes**, including compatibility with Net Metering, Net Billing/Net Feed-in, Virtual Net Metering, and Group Net Metering (referencing DERC and Kerala provisions).
- Open Access applicability, where relevant for multi-licensee areas.
- P2P pilots, sandbox mechanisms, or **phased rollouts** approved by the Distribution Licensee

Objectives of the P2P Energy Trading Regulation

- Facilitate the **integration of distributed renewable energy resources** into the electricity distribution system in a structured and transparent manner.
- **Promote rooftop solar** adoption by enabling prosumers to monetize surplus renewable energy.
- **Develop a secure and transparent digital marketplace** for renewable energy transactions using blockchain or equivalent technologies.
- **Enhance consumer participation** in electricity markets by enabling voluntary renewable energy procurement.
- **Support DISCOMs in optimizing grid operations**, improving visibility of distributed energy resources, and reducing technical losses.
- **Establish clear regulatory guidelines** for scheduling, metering, settlement, and transaction monitoring.
- **Provide a scalable framework** that can evolve into advanced energy systems such as Virtual Power Plants (VPP), demand response, and integrated distributed energy markets.

Guidelines for P2P Implementation

Voluntary Participation: Participation in P2P energy trading shall be voluntary for both prosumers and consumers.

Technology Neutrality: Blockchain is preferred, but other technologies offering equivalent transparency, security, and auditability may be approved.

Transparency and Traceability: All energy transactions shall be recorded in verifiable and tamper-proof digital ledgers to ensure auditability.

Grid Safety and Technical Compliance: All P2P transactions must comply with technical standards prescribed by CEA, UPERC, and the Distribution Licensee.

Cybersecurity and Data Privacy: P2P platforms shall comply with national cybersecurity standards and ensure protection of consumer data.

Non-Discriminatory Market Access: All eligible participants shall have equal opportunity to participate without discriminatory practices.

DISCOM-Centric Grid Control: Distribution Licensees shall retain operational control of the grid to ensure system stability and safety.

Participation Framework

Eligible P2P Prosumers

Consumers with approved rooftop solar systems connected to the distribution network and equipped with smart meters.

Eligible P2P Consumers

Consumers with smart meters who wish to purchase renewable energy through the P2P platform.

Eligible Service Providers

Entities capable of operating digital P2P platforms with blockchain or equivalent technology and integrating with DISCOM systems.

Distribution Licensee

Responsible for network feasibility evaluation, meter data validation, energy accounting, and settlement.

Registration Process

Participants shall register with the Distribution Licensee and the authorized Service Provider before participating in P2P transactions.

Switching Between Schemes

Participants may switch between P2P and other rooftop solar schemes (such as Net Metering or Net Billing) once per financial year as per regulatory provisions.

Roles and Responsibilities

Prosumers:

- Right to sell surplus energy at mutually agreed price
- Access to transparent digital logs and billing
- Must submit schedules, maintain RTS, and avoid over-injection

Consumers:

- Right to buy renewable energy at negotiated prices
- Must maintain prepaid e-Wallet and adhere to schedules

DISCOMs:

- Ensure system safety, data validation, metering, and integration
- May curtail for network safety, DT overload, voltage issues

Service Provider:

- Ensure platform uptime, neutrality, cybersecurity, and API integration
- Provide dashboards, reports, and daily settlements

Technical Standards for P2P Integration

Smart Metering Infrastructure

All P2P participants must have smart meters capable of recording import and export energy in 15-minute time blocks.

Meter Data Management (MDM)

All settlement and energy accounting shall rely on validated meter data obtained from the DISCOM's Meter Data Management system.

Communication Infrastructure

Meter communication shall comply with DLMS/COSEM and other CEA-approved protocols.

Data Integration

The P2P platform shall access validated meter data through secure APIs provided by the Distribution Licensee.

System Stability

P2P transactions shall not adversely impact network parameters including voltage, frequency, transformer loading, and power quality.

P2P Platform Architecture and Cybersecurity

Digital Transaction Platform

The P2P platform shall provide matching, scheduling, and settlement functionalities.

Blockchain / Distributed Ledger

Transaction records shall be stored in tamper-proof digital ledgers ensuring transparency and auditability.

Integration with DISCOM Systems

Platforms must integrate with:

MDM systems
Billing systems
Consumer databases

Cybersecurity Compliance

Platforms must comply with:

CERT-In guidelines
NCIIPC critical infrastructure guidelines
National data privacy frameworks

Data Residency

All consumer data and transaction records must be stored within India.

Transaction Scheduling and Settlements

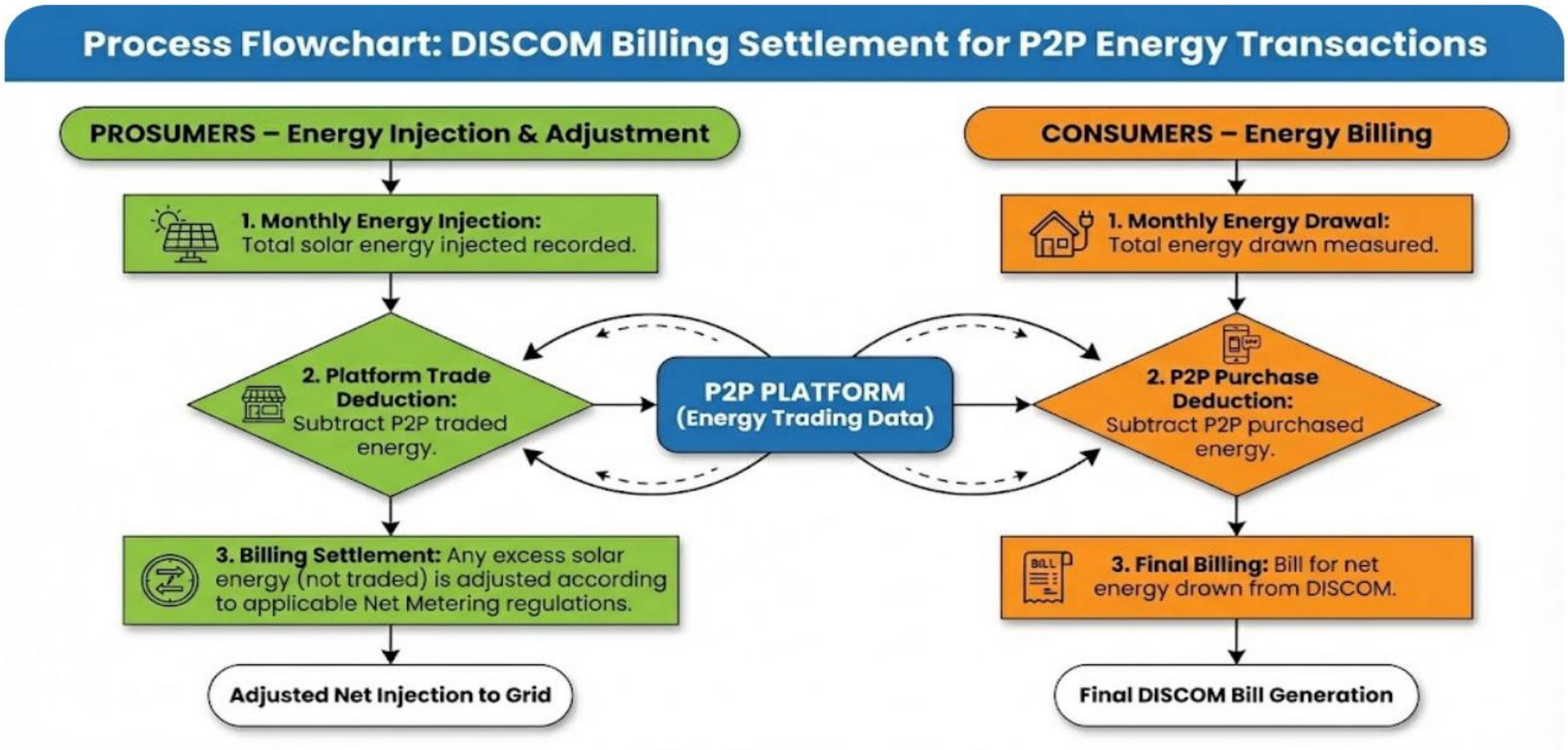
Scheduling (Day-Ahead Scheduling - DAS)

- Mandatory Requirement: All P2P Transactions must be based on Day-Ahead Schedules.
- Submission Timeline (Day N-1): P2P Participants must submit their schedule for the next day (Day N) by 17:00 hours.
- Finalization: The Service Provider prepares the P2P Schedule, and the Distribution Licensee completes feasibility confirmation by 20:00 hours of the previous day (N-1).
- No Deviation Allowed: No deviation in the schedule is allowed beyond the submission time.

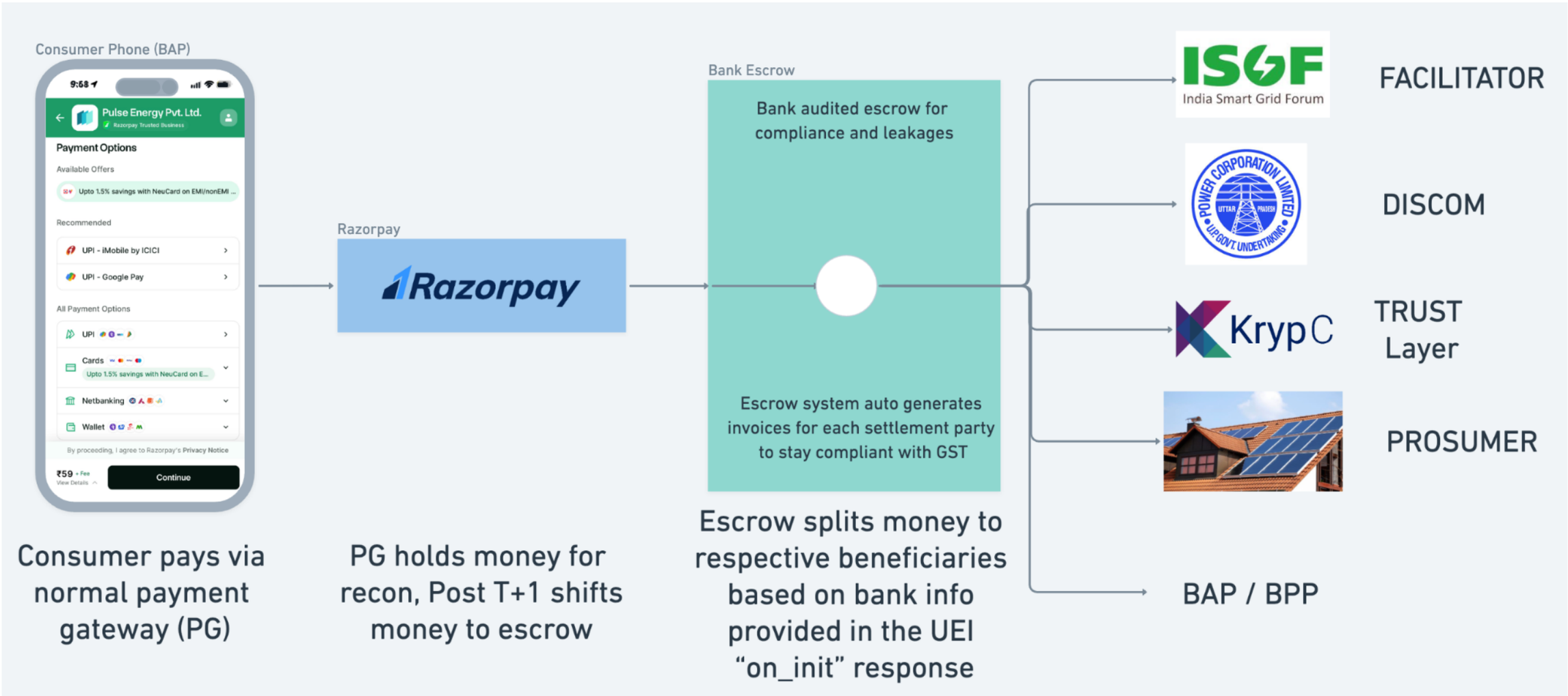
Billing and Settlement

- P2P Billing Cycle: The P2P Platform's billing cycle shall be daily
- Settlement Basis: Settlement uses time block-wise meter data fetched by the Service Provider from the DISCOM's MDM system.
- Payment Flow: P2P Consumers pay via the pre-paid eWallet.
- Charges: The Transaction Price is mutually agreed. Other charges (Wheeling, CSS, etc.) apply as determined by SERC.

DISCOM Billing Settlement for P2P Energy Transactions



Payment and Settlement for P2P Energy Trading Pilot



Implementation Roadmap for P2P Energy Trading

Phase 1: Pilot Deployment

- Initial deployment in selected DISCOM service areas with high smart meter penetration and rooftop solar installations.
- Testing of platform integration with DISCOM systems including smart meters, MDM systems, and billing platforms.
- Participation limited to a defined number of prosumers and consumers to evaluate operational performance.

Phase 2: Controlled Expansion

- Gradual increase in the number of participants across urban and semi-urban areas.
- Integration with additional rooftop solar installations and distributed renewable energy resources.
- Enhancement of digital platform capabilities including automated scheduling, settlement, and reporting dashboards.

Phase 3: State-wide Deployment

- Expansion of P2P participation across all DISCOMs in Uttar Pradesh.
- Integration with advanced grid services including distributed energy management and local energy balancing.
- Development of interoperable digital platforms supporting large-scale P2P participation.

Phase 4: Advanced Energy Market Integration

- Integration with emerging frameworks such as:
- Virtual Power Plants (VPP)
- The India Energy Stack (IES)
- Demand Response Programs
- Battery Energy Storage Systems (BESS)
- Vehicle-to-Grid (V2G) services

Suggestions for SERCs

Regularity Certainty needed for the new P2P market to grow necessitates that:

- Regulations once notified should not be changed frequently; should refrain from amendments for 5-10 years
- Service Provider Charges should remain fixed for ~ 5 years
- Setting of Wheeling Charges and Cross Subsidy Surcharges should coincide with MYT period and be specified in Multi-Year Tariff Order in future

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